

Edwin Lock

Postdoctoral researcher
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Research interests

Algorithms and Complexity, Algorithmic Game Theory, Auctions and Markets

Research positions

- 2021-present **Postdoctoral Researcher**, *Computer Science and Economics Departments, Oxford University.*
2021–2024 **Research Fellow**, *Nuffield College, Oxford University.*

Education

- 2017–2022 **DPhil in Computer Science**, *Oxford University.*
Advisor: PW Goldberg. Supported by an EPSRC Scholarship.
2016–2017 **MSc in Mathematics and Foundations of Computer Science**, *Oxford University.*
2013–2016 **BSc in Mathematics**, *FernUniversität in Hagen, Germany.*
2008–2012 **BA in Music**, *Oxford University.*

Journal papers

- R & R Competitive and Optimal Pricing with Budgets.
S Finster, PW Goldberg, and E Lock.
Theoretical Economics.
Revise and Resubmit.
- 2023 Solving Strong-Substitutes Product-Mix Auctions.
E Baldwin, PW Goldberg, P Klemperer, E Lock.
Mathematics of Operations Research.
- 2022 Learning Strong Substitutes Demand via Queries.
PW Goldberg, E Lock and F Marmolejo-Cossío.
Transactions on Economics and Computation.
- 2019 Characterising and Recognising Game-Perfect Graphs.
SD Andres and E Lock.
Discrete Mathematics and Theoretical Computer Science.

Conference papers

- 2024 Decentralized Convergence to Equilibrium Prices in Trading Networks.
E Lock, BP Evans, E Kreacic, S Bhatt, A Koppel, S Ganesh, PW Goldberg.
39th Conference on Artificial Intelligence (AAAI'25)
- 2024 The Computational Complexity of the Housing Market.
E Lock, Z Qiu, A Teytelboym.
17th International Symposium on Algorithmic Game Theory (SAGT'24).
- 2023 Substitutes markets with budget constraints: solving for competitive and optimal prices.
S Finster, PW Goldberg, and E Lock.
19th Conference on Web and Internet Economics (WINE'23).
- 2023 Welfare-Maximizing Pooled Testing.
S Finster, M González Amador, E Lock, F Marmolejo Cossío, E Micha, AD Procaccia.
24th ACM Conference on Economics and Computation (EC'23).
Awarded best paper in applied modelling track.
Invited seminar at the Institute for Societal Computing, Saarbrücken.
- 2021 Optimal Testing and Containment Strategies for Universities in Mexico amid COVID-19.
L Benavides-Vázquez, H A Guzmán-Gutiérrez, J Jonnerby, P Lazos, E Lock, F J Marmolejo-Cossío, N Rajgopal and JR Tello-Ayala.
1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO'21).

- 2020 Learning Strong Substitutes Demand via Queries.
PW Goldberg, E Lock and F Marmolejo-Cossio.
16th Conference on Web and Internet Economics (WINE'20).
Nominated for best paper.
- 2020 Test and Contain: A Resource-Optimal Testing Strategy for COVID-19.
J Jonnerby, P Lazos, E Lock, F Marmolejo-Cossio, CB Ramsey and D Sridhar.
AI for Social Good 2020, Harvard CRCS Workshop.
Best poster video award at the *ACM SIGecom Global Challenges for Economics and Computation workshop* at EC'20.

Working papers

Implementing Walrasian Equilibrium: the Languages of Product-mix Auctions.
E Baldwin, P Klemperer, E Lock.
SSRN preprint.
Invited seminar at the Center for Research in Economics and Statistics (CREST), Paris.

Accelerated Preference Elicitation with LLM-Based Proxies.
D Huang, E Lock, F Marmolejo-Cossio, D Parkes.

Teaching and Supervision

- 2018–2021 **Departmental teacher**, *Computer Science Department, Oxford University*.
Taught postgraduate course in Computational Complexity to MSc and fourth-year undergraduate students. Supervised MSc theses, and demonstrated practicals.
- 2020–2022 **Lecturer**, *Balliol College, Oxford University*.
Taught undergraduate students in a range of computer science and mathematics courses. Set examinations and assisted with undergraduate admissions. (Also taught for six other Oxford Colleges.)
- 2019–now **Other supervision**.
Currently supervising undergraduate projects at Harvard University. Past supervision of postdoc hired under ACM GCEC'20 grant, and student interns (from Harvard and Edinburgh Universities) as part of *Test and Contain* research project.
- 2019 **Organiser and tutor**, *Dr. HN Science Centre, Gauribidanur, India*.
Organised, ran and taught at a science workshop in rural India. Introduced approx. 200 students to scientific techniques, provided hands-on experience with scientific equipment and familiarised students with cutting-edge research topics.

Professional activities

- 2021–now **Program Committee member** for conferences including *Economics and Computation (EC)*, the *Conference On Web And InterNet Economics (WINE)* and *Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)*.
- 2021–now **Reviewer** for journals including *Naval Research Logistics*, *Mathematics of Operations Research*, and *Information Processing Letters*.
- 2021 **Co-organised tutorial** on "Fairness and Discrimination through the Dual Lens of Mechanism Design and Machine Learning" at EC'21.
- 2020–2023 **Founding member and principal organiser** of the *Mechanism Design for Social Good (MD4SG)* healthcare working group. Supervised several research projects allocated between 12 group members.

Projects

- COVID-19 Testing I co-founded Test and Contain, a project to design and implement resource-optimal COVID-19 testing. **Supported by an ACM SIGecom research grant (\$25,000)**. Initiated and led collaboration with the Institute for Scientific Research and Technology (IPICYT) in Mexico on experimental trial to pilot the testing mechanism and algorithms in our paper "Welfare-Maximizing Pooled Testing". See www.c-sef.com for details.
- Debt restructuring Collaborated with staff of the International Monetary Fund (IMF) to design an auction-based sovereign debt restructuring mechanism.
- Tax credit markets Worked with industry partner, Bellus Ventures, to design a market platform for developers of renewable energy projects to sell Investment Tax Credits (ITCs).